

Assignment 1 - VaR and ES

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1 DATA AND PORTFOLIO

1.1 Assets and Sources

The portfolio consists of five risky assets, a foreign exchange rate for EUR/USD and a EURIBOR 3-month rate. Equity and FX data are sourced from Yahoo Finance via the `yfinance` API; EURIBOR 3-month rates are obtained from a separate CSV file of ECB published rates. All data is on a daily frequency.

Ticker	Asset	Type	Currency
ASML	ASML Holding N.V.	Equity	EUR
SHELL	Shell PLC	Equity	EUR
JPM	JPMorgan Chase & Co.	Equity	USD [†]
STOXX50	Euro Stoxx 50	Index	EUR
SP500	S&P 500	Index	USD [†]
EURUSD	EUR/USD exchange rate	FX	—
EURIBOR3M	3-month EURIBOR	Interest Rate	—

Table 1.1: Assets and data sources. [†] USD-denominated assets are EUR-adjusted in the returns series.

The full dataset spans January 2015 to March 2026. The sample is split into a pre-sample period used exclusively for portfolio optimization, for which the first 2 years are used. The period used for VaR and ES testing is therefore from 01-01-2017 to 31-03-2026.

1.2 Data Cleaning

To deal with missing observations, we first check if multiple observations are missing on the same day. For days where 4 or more assets are missing, we remove the date entirely — a total of 50 out of 2928 days. An inspection shows that days with 6 missing assets are global market closures, only one date had 5 missing assets and was an Easter Monday, and dates with 4 missing assets were European holidays, as the data consists of 4 European listed assets.

Data is checked for consecutive gaps, and is forward-filled, with a single backfill for a missing observation in the first row. To check if this filling process leads to abnormally

high returns, overlap is checked with days where the return exceeded 7.5%, with no overlap out of 56 such days. The cleaned dataset of prices is then stored.

1.3 Return Construction

EUR ADJUSTMENT FOR USD-DENOMINATED ASSETS. JPM and the S&P 500 index are priced in USD. Prices are converted to EUR using the spot rate:

$$P_{i,t}^{\text{EUR}} = \frac{P_{i,t}^{\text{USD}}}{S_t}$$

where S_t is the EUR/USD rate (USD per EUR). Prices and returns in USD terms are also present in the dataset for later analysis of currency shocks. We therefore consider the portfolio from the perspective of a European investor.

LOG-RETURNS. Continuously compounded log-returns are computed for all asset price series:

$$r_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right)$$

Log-returns are preferred over simple returns for three reasons — they are additive, approximately normally distributed for daily frequencies and are symmetric around zero.

EURIBOR 3-MONTH RATE. Unlike equity prices, interest rates are not strictly positive and their level, not their changes, drives bond P&L. The EURIBOR 3-month rate therefore enters as a daily first difference:

$$\Delta y_t = y_t - y_{t-1}$$

The loan's mark-to-market can then be used to study portfolio P&L.

1.4 Portfolio Construction

SEPARATION. The portfolio is separated: total wealth $W = \text{€}1,000,000$ is split into a risky portfolio (80%) and a risk-free position (20%). The 20% position is modeled as a fixed-rate EUR loan rather than a cash deposit, introducing interest-rate sensitivity as an additional risk factor alongside equity and currency risk.

TANGENCY PORTFOLIO. Within the 80% risky sleeve, weights are determined by maximizing the Sharpe ratio using the pre-sample covariance matrix $\hat{\Sigma}$ and pre-sample mean returns $\hat{\mu}$:

$$\max_w \frac{w^\top \hat{\mu} - r_f}{\sqrt{w^\top \hat{\Sigma} w}} \quad \text{s.t.} \quad \mathbf{1}^\top w = 1, \quad 0.10 \leq w_i \leq 0.30 \quad \forall i$$

There are possible issues with simply holding a mean-variance optimized portfolio, optimized on only 6 assets and 2 years of data. One such issue is time variance in variances and covariances, but as this is a risk management assignment this approach was deemed adequate. Weights are kept constant throughout testing, and hence the portfolio is rebalanced daily.

Individual weight bounds of [0.10, 0.30] for the risky portfolio are used so the portfolio is not too concentrated on any individual asset. The final portfolio weights—already scaled by the 0.80 risky fraction—are reported in Table 1.2.

Asset	Weight	Notional (€)
ASML	16.00%	160,000
SHELL	8.00%	80,000
JPM (EUR-adj.)	24.00%	240,000
EURO STOXX 50	8.00%	80,000
S&P 500 (EUR-adj.)	24.00%	240,000
Loan (fixed-rate)	20.00%	200,000
Total	100%	1,000,000

Table 1.2: Portfolio weights and allocation assuming a notional of €1,000,000 and US assets converted to Euro.

FIXED-RATE LOAN. The loan position is modelled as a fixed-rate EUR bond with the following parameters. The coupon is locked at the prevailing EURIBOR 3-month rate on 3 January 2017 plus a 150 bps credit spread. We keep the loan as a fixed loan with a long duration so it remains more sensitive to interest rate risk to make analysis more interesting:

$$c = r_f + 150 \text{ bps} = -0.319\% + 1.50\% = 1.181\% \text{ p.a.}$$

The maturity is the length of the main sample ($T = 9.243$ years), and modified duration is approximated as:

$$D^* = \frac{T}{1 + c} = \frac{9.243}{1.01181} \approx 9.135 \text{ years}$$

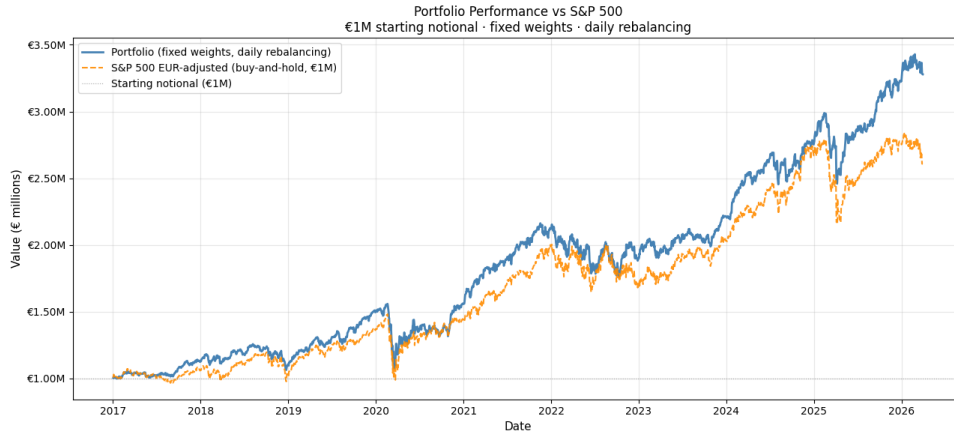


Figure 1.1: Portfolio performance, daily readjusted with fixed weights.

Portfolio performance is reported in Figure 1.1. As expected, the portfolio and S&P index appear strongly correlated. The appendix also includes graphs of the price and return series for assets and the efficient frontier.

2 METHODS

2.1 Model 1: Multivariate Normal

2.1.1 Method

Under normality the 1-day VaR and ES at confidence level $1 - \alpha$ are

$$\text{VaR}_\alpha = -\mu_p - \sigma_p \Phi^{-1}(\alpha), \quad (2.1)$$

$$\text{ES}_\alpha = -\mu_p + \sigma_p \frac{\phi(\Phi^{-1}(\alpha))}{\alpha}, \quad (2.2)$$

where Φ^{-1} and ϕ are the standard normal quantile and density functions.

2.1.2 Results

Table 2.1 reports the 1-day VaR and ES at $\alpha = 1\%$ and 5% for each asset and the portfolio. The portfolio VaR is substantially lower than the sum of individual VaRs, reflecting the diversification benefit captured by the off-diagonal entries of Σ .

Table 2.1: 1-day VaR and ES under the Normal model (EUR).

Entity	VaR		ES	
	$\alpha = 1\%$	$\alpha = 5\%$	$\alpha = 1\%$	$\alpha = 5\%$
ASML	7 854	5 494	9 028	6 941
Shell	3 296	2 317	3 782	2 917
JPM (EUR)	9 842	6 907	11 301	8 707
Stoxx 50	2 082	1 465	2 388	1 843
S&P 500 (EUR)	6 745	4 736	7 745	5 968
Loan	679	483	777	603
Portfolio	22 621	15 833	25 996	19 995

2.1.3 Backtest

We backtest the 1% VaR using an expanding-window scheme: on each day t , μ and Σ are re-estimated from all data up to $t - 1$ (minimum 250 observations), and a violation is recorded when the realised loss $L_t = -V w^\top R_t$ exceeds VaR_t . Under correct specification the expected violation rate equals α , i.e. roughly 2.55 per year given approximately 255 trading days.

Table 2.2 reports violations by year. The model produces 46 violations over 2 366 days (hit rate 1.94%, nearly double the nominal 1%), with a severe concentration in 2020 (19 violations against 2.57 expected) driven by the COVID shock.

Table 2.3 compares the average predicted ES against the average realised loss on violation days by year. The model underestimates most severely in 2020, where the average realised loss (EUR,40,824) exceeds the predicted ES (EUR,24,026) by 70%, driven by the COVID shock. A similar gap appears in 2025.

Table 2.2: Normal model yearly VaR violations at $\alpha = 1\%$.

Year	Violations	Trading days	Expected
2017	1	256	2.56
2018	7	255	2.55
2019	3	255	2.55
2020	19	257	2.57
2021	1	258	2.58
2022	7	257	2.57
2023	1	255	2.55
2024	3	256	2.56
2025	4	255	2.55
2026	0	62	0.62
Total	46	2 366	23.66

Table 2.3: Normal model — average ES vs realised shortfall on violation days (EUR).

Year	Violations	Avg predicted ES	Avg realised loss
2017	1	23 459	22 816
2018	7	21 507	23 015
2019	3	21 241	23 014
2020	19	24 026	40 824
2021	1	26 633	27 594
2022	7	27 106	27 100
2023	1	27 443	25 559
2024	3	26 164	29 477
2025	4	26 015	38 886

2.2 Model 2: Multivariate Student- t

2.2.1 Method

To accommodate heavier tails we assume standardised residuals follow a Student- t distribution with ν degrees of freedom. The unit-variance scale adjustment $s = \sigma_p \sqrt{(\nu - 2)/\nu}$ is necessary because a raw Student- t with ν degrees of freedom has variance $\nu/(\nu - 2)$, not unity; without this correction the distribution would be too wide and σ_p would no longer represent the true standard deviation. The risk measures become

$$\text{VaR}_\alpha = -\mu_p - s t_\nu^{-1}(\alpha), \quad (2.3)$$

$$\text{ES}_\alpha = -\mu_p + s \frac{\nu + [t_\nu^{-1}(\alpha)]^2}{(\nu - 1)\alpha} f_\nu(t_\nu^{-1}(\alpha)), \quad (2.4)$$

where t_ν^{-1} and f_ν are the Student- t quantile and density.

2.2.2 Selecting the Degrees of Freedom

We pool the standardised residuals of all six assets over the pre-sample and compare against the theoretical quantiles of the unit-variance Student- t for $\nu \in \{3, 4, 5, 6\}$. The

candidate minimising the QQ-plot RMSE is selected. Table 2.4 and Figure 2.1 show that $\nu = 5$ achieves the lowest RMSE (0.0544) and is used throughout.

Table 2.4: QQ-plot RMSE for candidate degrees of freedom (pre-sample).

ν	QQ-RMSE
5	0.0544
6	0.0717
4	0.0750
3	0.1877

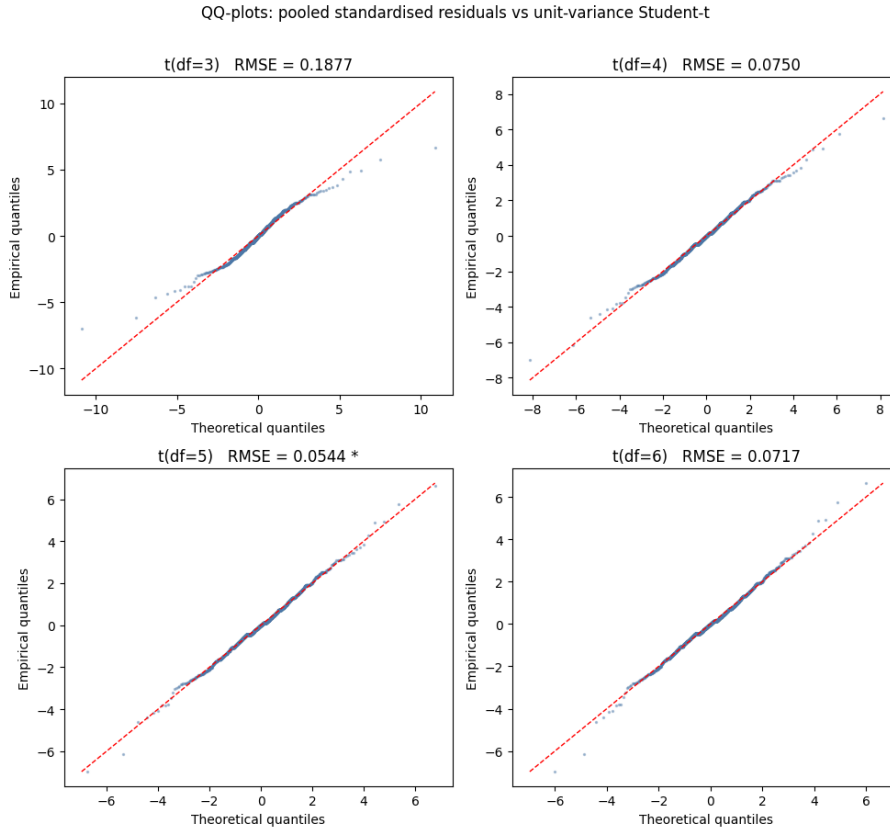


Figure 2.1: QQ-plots of pooled pre-sample standardised residuals.

2.2.3 Results

Table 2.5 reports VaR and ES under the Student- t model. Compared to the Normal, the 1% VaR increases by roughly 12% at the portfolio level (EUR 25 411 vs 22 621) and the 1% ES by 30% (EUR 33 802 vs 25 996), reflecting the heavier tails of the t -distribution.

Table 2.6 shows the same comparison. The gap narrows in most years, though 2020 remains problematic (EUR,44,259 realised vs EUR,31,072 predicted). In non-crisis years such as 2018, 2019, and 2022 the predicted ES actually exceeds the realised loss, indicating adequate tail coverage outside of stress periods.

Table 2.5: 1-day VaR and ES under the Student- t ($\nu = 5$) model (EUR).

Entity	VaR		ES	
	$\alpha = 1\%$	$\alpha = 5\%$	$\alpha = 1\%$	$\alpha = 5\%$
ASML	8 824	5 203	11 741	7 550
Shell	3 698	2 197	4 908	3 170
JPM (EUR)	11 048	6 545	14 675	9 464
Stoxx 50	2 335	1 389	3 097	2 002
S&P 500 (EUR)	7 572	4 488	10 056	6 487
Loan	760	459	1 002	654
Portfolio	25 411	14 996	33 802	21 748

Table 2.6: Student- t ($\nu = 5$) model — average ES vs realised shortfall on violation days (EUR).

Year	Violations	Avg predicted ES	Avg realised loss
2018	4	27 978	25 185
2019	2	27 624	24 560
2020	16	31 072	44 259
2021	1	34 649	27 594
2022	4	35 441	28 532
2024	2	33 995	32 795
2025	4	33 821	38 886

2.2.4 Backtest

The same expanding-window procedure is applied. Table 2.7 shows the Student- t model yields 33 violations (hit rate 1.39%), closer to the nominal 1% than the Normal's 1.94%. The 2020 concentration remains (16 violations) but is reduced from 19.

Table 2.7: Student- t ($\nu = 5$) model — yearly VaR violations at $\alpha = 1\%$.

Year	Violations	Trading days	Expected
2017	0	256	2.56
2018	4	255	2.55
2019	2	255	2.55
2020	16	257	2.57
2021	1	258	2.58
2022	4	257	2.57
2023	0	255	2.55
2024	2	256	2.56
2025	4	255	2.55
2026	0	62	0.62
Total	33	2 366	23.66

2.3 Model Comparison and Diagnostics

Table 2.8 summarises the backtest performance side by side.

Table 2.8: Backtest summary at $\alpha = 1\%$.

Model	Obs	Hits	Hit rate	Avg VaR (hits)	Avg loss (hits)
Normal	2 366	46	1.94%	21 201	32 944
Student- t ($\nu = 5$)	2 366	33	1.39%	23 790	36 996

Figure 2.2 plots realised losses against the VaR threshold; black dots mark violations. The Normal panel shows visibly more breaches, concentrated during the COVID stress period and the April 2025 shock.

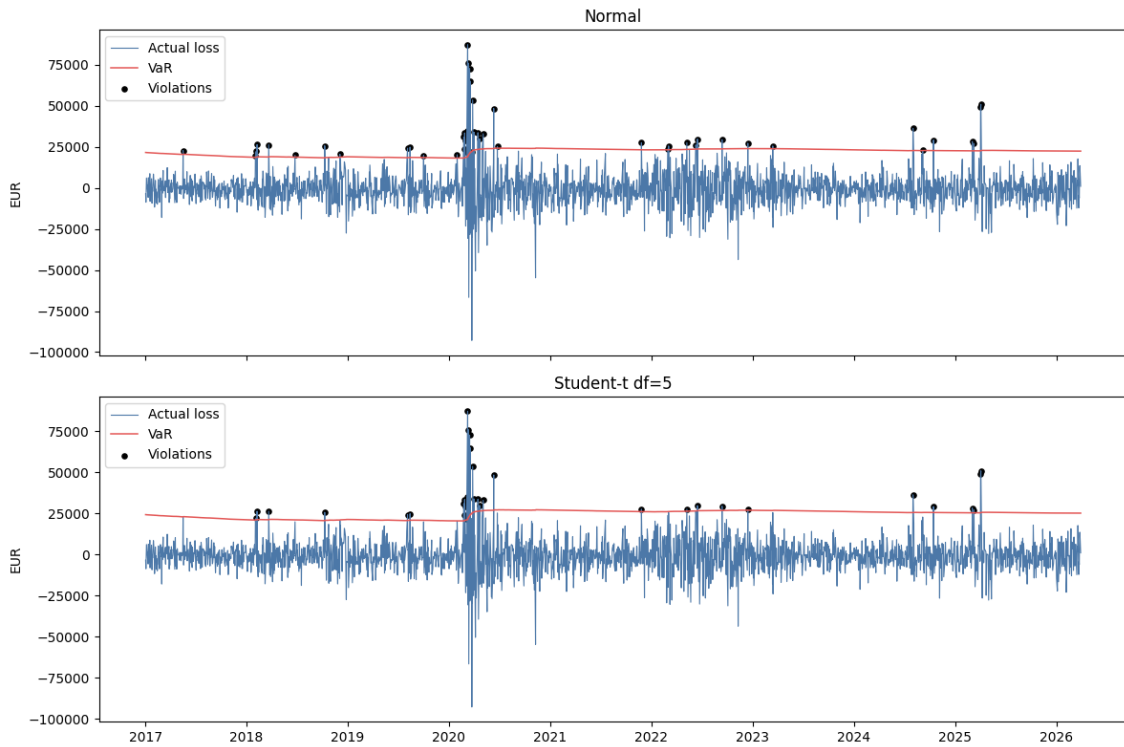


Figure 2.2: Realised loss against 1% VaR.

Figure 2.3 shows the rolling 60-day violation count. Under independence this should fluctuate around $60 \times 0.01 = 0.6$. Both models exhibit clear clustering around 2020 (peaking at 15–16 violations within a single 60-day window), indicating that the unconditional variance–covariance framework does not fully capture time-varying volatility. The Student- t reduces but does not eliminate this clustering, suggesting a conditional volatility model (e.g. GARCH) as a natural extension.

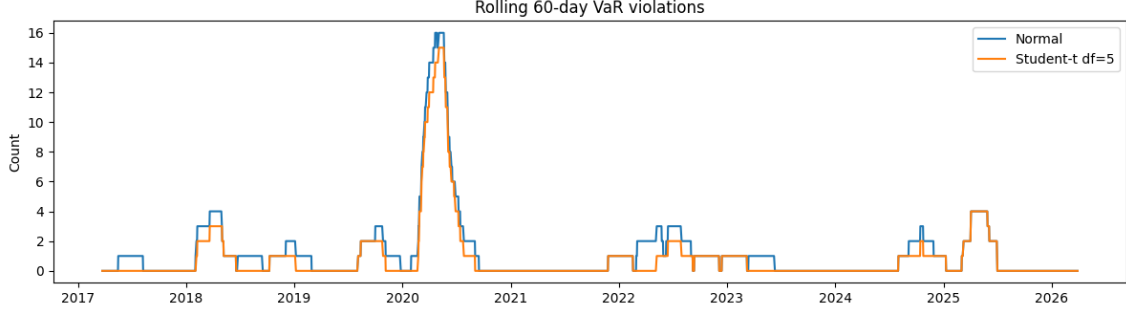


Figure 2.3: Rolling 60-day VaR violation count.

2.3.1 Effect of Sample Choice

To assess how the 2025 and early-2026 stress episodes influence the estimates, we compare portfolio risk measures estimated on data up to December 2024 against the full sample through March 2026 (Table 2.9).

Table 2.9: Portfolio 1% VaR and ES: sample sensitivity (EUR).

Sample	n	Normal		Student- t ($\nu = 5$)	
		VaR	ES	VaR	ES
Up to 2024/12	2 049	22 924	26 342	25 750	34 248
Full (to 2026/3)	2 366	22 621	25 996	25 411	33 802

The differences are modest: the full sample VaR is marginally lower because the additional 317 observations include a mix of calm and volatile days, slightly diluting the per-observation volatility. This confirms that with an already long sample (2 049+ days), individual stress episodes have a limited impact on unconditional estimates. Our expanding-window backtest handles this naturally, parameters incorporate new information as it arrives, so the VaR threshold adjusts upward during stress and gradually recedes afterward.

2.4 Model 3: Historical Simulation

2.4.1 Method

Historical simulation (HS) estimates the loss distribution non-parametrically from observed data. For a portfolio with loss operator $l_{[t]}$, we construct a set of historically simulated losses by applying the current portfolio mapping to each past observation of risk-factor changes:

$$\tilde{L}_s = l_{[t]}(X_s), \quad s = t - n + 1, \dots, t.$$

VaR and ES are then estimated directly from the empirical distribution of $\{\tilde{L}_s\}$ using the L-estimators of McNeil, Frey, and Embrechts (2015, §9.2.6). With losses sorted in

ascending order $L_{(1)} \leq \dots \leq L_{(n)}$ and $K = \lceil n\alpha \rceil$:

$$\widehat{\text{VaR}}_\alpha = L_{(K)}, \quad (2.5)$$

$$\widehat{\text{ES}}_\alpha = \frac{1}{n(1-\alpha)} \left[\sum_{k=K+1}^n L_{(k)} + (K - n\alpha) L_{(K)} \right]. \quad (2.6)$$

The fractional boundary weight $(K - n\alpha)$ on $L_{(K)}$ ensures consistency with the integral definition of ES; the simpler tail mean is a special case when $n\alpha$ is integer. For a linear portfolio the historically simulated losses reduce to $\tilde{L}_s = -V w^\top X_s$, so no delta or gamma approximation is required.

2.4.2 Results

Table 2.10 reports the 1-day VaR and ES at $\alpha = 1\%$ and $\alpha = 2.5\%$ using the full sample as estimation window ($n = 2,366$ days).

Table 2.10: 1-day VaR and ES under Historical Simulation (EUR).

Entity	VaR		ES	
	$\alpha = 1\%$	$\alpha = 2.5\%$	$\alpha = 1\%$	$\alpha = 2.5\%$
ASML	9 489	7 142	13 115	10 013
Shell	4 168	2 867	6 394	4 613
JPM (EUR)	11 674	8 301	17 657	12 998
STOXX 50	2 929	1 809	3 922	2 943
S&P 500 (EUR)	8 107	6 170	12 455	9 205
Loan	941	648	1 263	973
Portfolio	27 799	20 134	43 440	31 305

The portfolio VaR at $\alpha = 1\%$ (EUR 27 799) is around 23% higher than the Normal model (EUR 22 621) and 9% higher than the Student- t model (EUR 25 411). The difference is even more pronounced for ES: the HS estimate of EUR 43 440 exceeds the Normal by 67% and the Student- t by 28%. This reflects that the historical record includes realised tail events such as the COVID-19 drawdown in 2020 and the April 2025 market shock; parametric models fitted to the full sample average over these episodes and produce smoother estimates. The diversification benefit remains evident: the portfolio VaR (EUR 27 799) is well below the sum of individual asset VaRs (EUR 37 308).

2.4.3 Backtest

The same expanding-window scheme used for Models 1 and 2 is applied: on each day t from observation 250 onwards, VaR and ES are re-estimated from all data up to t , and the violation indicator $\hat{I}_{t+1} = \mathbf{1}\{L_{t+1} > \widehat{\text{VaR}}_t^\alpha\}$ is recorded for day $t + 1$.

Table 2.11 reports the violation count, the binomial score statistic

$$Z = \frac{M - m(1 - \alpha)}{\sqrt{m\alpha(1 - \alpha)}},$$

and the associated p -value across both confidence levels over 2 120 backtest days.

Table 2.11: HS backtest: VaR violation counts and binomial score test.

α	Expected	Actual	Hit rate	Z	p -value
1%	21.2	27	1.27%	1.276	0.202
2.5%	52.9	67	3.16%	1.963	0.050

Table 2.12: Historical Simulation — yearly VaR violations at 1% and 2.5%.

Year	T	1%		2.5%	
		Expected	Actual	Expected	Actual
2017	6	0.06	0	0.15	0
2018	255	2.55	5	6.38	17
2019	255	2.55	2	6.38	5
2020	257	2.57	13	6.43	22
2021	258	2.58	0	6.45	2
2022	257	2.57	2	6.43	12
2023	255	2.55	0	6.38	1
2024	256	2.56	2	6.40	4
2025	255	2.55	3	6.38	4
2026	62	0.62	0	1.55	0
Total	2,120	21.20	27	52.93	67

Both tests pass at the 5% level, though the observed counts exceed the expected values in both cases—the model tends to slightly underestimate VaR. The 1% hit rate of 1.27% compares favourably to the Normal (1.94%) and Student- t (1.39%) models, confirming that using the empirical loss distribution better captures the tails present in the sample. The top row of Figure 2.4 illustrates the main weakness of unconditional HS: the VaR line moves slowly upward as extreme observations accumulate in the expanding window, but it does not spike during the 2020 stress period. Violations therefore cluster in 2020 rather than being spread evenly over time.

The ES backtest uses the violation residual $\hat{K}_{t+1} = [(L_{t+1} - \widehat{\text{ES}}_t^\alpha) / \widehat{\text{ES}}_t^\alpha] \hat{I}_{t+1}$ with $\hat{\mu} = 0$; under correct specification the non-zero residuals should have mean zero. Table 2.13 reports the mean violation residual and the result of a one-sample t -test.

Table 2.13: HS backtest: ES violation residual t -test.

α	n_{viol}	$\bar{K}$	t -stat	p -value
1%	27	0.153	1.493	0.148
2.5%	67	0.072	1.044	0.300

Both levels pass the ES test. The positive mean residuals ($\bar{K} > 0$) indicate that on violation days realised losses do exceed the ES forecast on average, but the effect is not statistically significant. The elicibility score (Table 2.20) provides an additional comparison of VaR forecast accuracy across methods.

Table 2.14: HS — average ES vs. realised shortfall on violation days (EUR).

Year	$\alpha = 1\%$			$\alpha = 2.5\%$		
	Violations	Avg ES	Avg Loss	Violations	Avg ES	Avg Loss
2017	0	—	—	0	—	—
2018	5	21,112	24,560	17	18,544	18,530
2019	2	24,378	24,969	5	20,703	20,922
2020	13	40,381	50,045	22	31,236	39,639
2021	0	—	—	2	35,655	24,719
2022	2	48,970	30,306	12	34,485	25,332
2023	0	—	—	1	33,622	26,171
2024	2	43,643	34,628	4	31,560	28,386
2025	3	43,419	44,206	4	31,509	40,073
2026	0	—	—	0	—	—
Total/Avg	27	36,843	40,215	67	28,014	29,033

2.5 Model 4: Filtered Historical Simulation

2.5.1 Method

Standard HS is unconditional: the risk estimate does not respond to changes in current market volatility. Filtered Historical Simulation (FHS) addresses this by standardising historical returns by a time-varying volatility estimate before simulating future losses (McNeil, Frey, and Embrechts, 2015, §9.2.4).

We follow the multivariate EWMA approach. For each asset i , the conditional variance is updated recursively with decay parameter $\lambda = 0.94$ (RiskMetrics):

$$\hat{\sigma}_{t,i}^2 = \lambda \hat{\sigma}_{t-1,i}^2 + (1 - \lambda) r_{t-1,i}^2,$$

seeded with the sample variance. Standardised residuals $\hat{z}_{t,i} = r_{t,i}/\hat{\sigma}_{t,i}$ serve as iid proxies for the innovation distribution. The one-step-ahead volatility forecast is obtained by advancing the recursion one period:

$$\hat{\sigma}_{t+1,i}^2 = \lambda \hat{\sigma}_{t,i}^2 + (1 - \lambda) r_{t,i}^2.$$

Simulated risk-factor changes for tomorrow are constructed by rescaling each historical residual by tomorrow's forecast volatility (with $\hat{\mu} = 0$):

$$\tilde{X}_{s,i} = \hat{\sigma}_{t+1,i} \hat{z}_{s,i}, \quad s = t - n + 1, \dots, t.$$

The simulated portfolio losses $\tilde{L}_s = l_{[t]}(\tilde{X}_s)$ are then used with the same L-estimators as in Model 3 to obtain VaR and ES. This procedure preserves the empirical tail shape of the residuals while scaling the entire distribution to reflect today's estimated volatility level.

2.5.2 Results

Table 2.15 reports the FHS risk measures, estimated at end-of-sample ($t = \text{March 2026}$).

The FHS portfolio VaR at $\alpha = 1\%$ (EUR 28 053) is close to the HS estimate (EUR 27 799), reflecting that volatility at end-of-sample is near its historical average. The ES tells a

Table 2.15: 1-day VaR and ES under Filtered Historical Simulation / EWMA (EUR).

Entity	VaR		ES	
	$\alpha = 1\%$	$\alpha = 2.5\%$	$\alpha = 1\%$	$\alpha = 2.5\%$
ASML	11 753	8 495	16 334	12 543
Shell	4 009	2 947	5 306	4 165
JPM (EUR)	10 595	7 838	13 609	10 839
STOXX 50	3 140	2 429	4 181	3 306
S&P 500 (EUR)	8 231	6 048	10 734	8 465
Loan	975	733	1 362	1 031
Portfolio	28 053	20 981	35 499	28 536

different story: at EUR 35 499 it is 18% below the HS figure of EUR 43 440. By anchoring the simulation to current rather than average volatility, FHS produces a tighter conditional distribution; extreme historical residuals are downscaled when today’s volatility is moderate, reducing the average exceedance above VaR. At the asset level, ASML shows the largest shift—its FHS VaR exceeds the HS figure by 24%—indicating that its end-of-sample EWMA volatility is meaningfully above the full-sample average. JPM moves in the opposite direction, with FHS VaR below HS (EUR 10 595 vs EUR 11 674), consistent with USD-denominated exposure being relatively calm in early 2026.

2.5.3 Backtest

The expanding-window backtest for FHS uses the same 2 120-day evaluation period. At each step the EWMA recursion is applied to all data up to t , yielding the one-step-ahead forecast $\hat{\sigma}_{t+1}$; residuals $\hat{z}_s$ ($s \leq t$) are then rescaled to form the simulated loss distribution from which VaR and ES are read off.

Table 2.16: FHS-EWMA backtest: VaR violation counts and binomial score test.

α	Expected	Actual	Hit rate	Z	p -value	Pass
1%	21.2	19	0.90%	−0.472	0.637	Yes
2.5%	52.9	54	2.55%	0.153	0.878	Yes

FHS passes both binomial tests comfortably, with actual violations below the expected count at both levels. The Z -statistics are negative, indicating mild conservatism. The 1% hit rate of 0.90% is the lowest across all four models and is statistically indistinguishable from the nominal 1%. The bottom row of Figure 2.4 shows the key difference relative to HS: the FHS VaR line spikes sharply during the COVID stress of 2020 and the April 2025 shock, closely tracking the realised loss envelope. As a result, violations are spread more evenly over time rather than clustering at stress events.

The ES residuals are close to zero in both cases and the t -tests pass easily. The near-zero mean residuals suggest that on violation days the FHS ES forecast is well-calibrated: realised losses on exceedance days do not systematically exceed or undercut the conditional ES estimate.

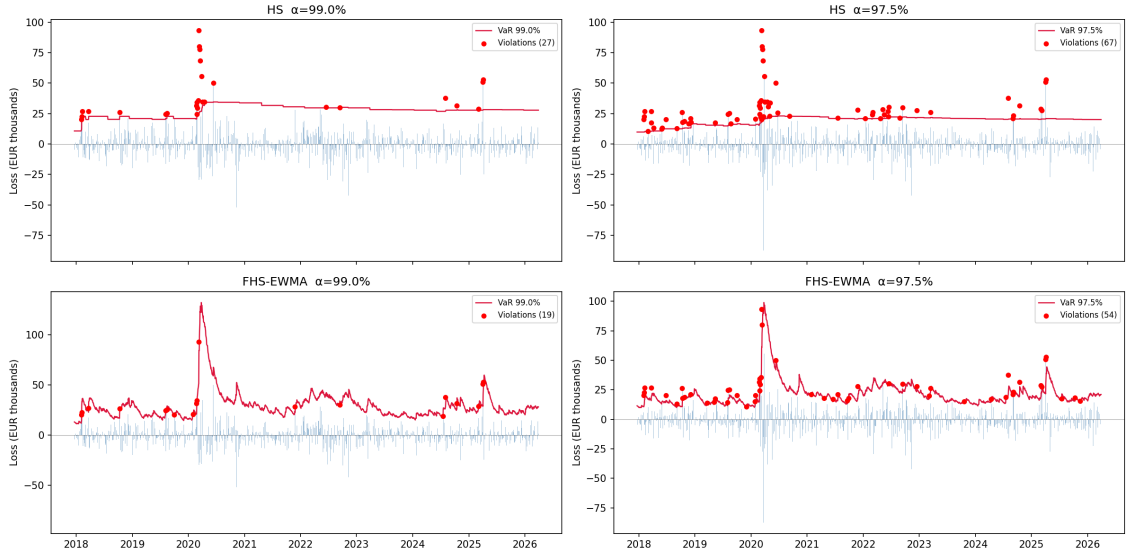


Figure 2.4: Expanding-window VaR backtest: realised portfolio losses (bars) against the estimated VaR threshold (line) for HS (top row) and FHS-EWMA (bottom row) at $\alpha = 1\%$ (left) and $\alpha = 2.5\%$ (right). Red dots mark violation days.

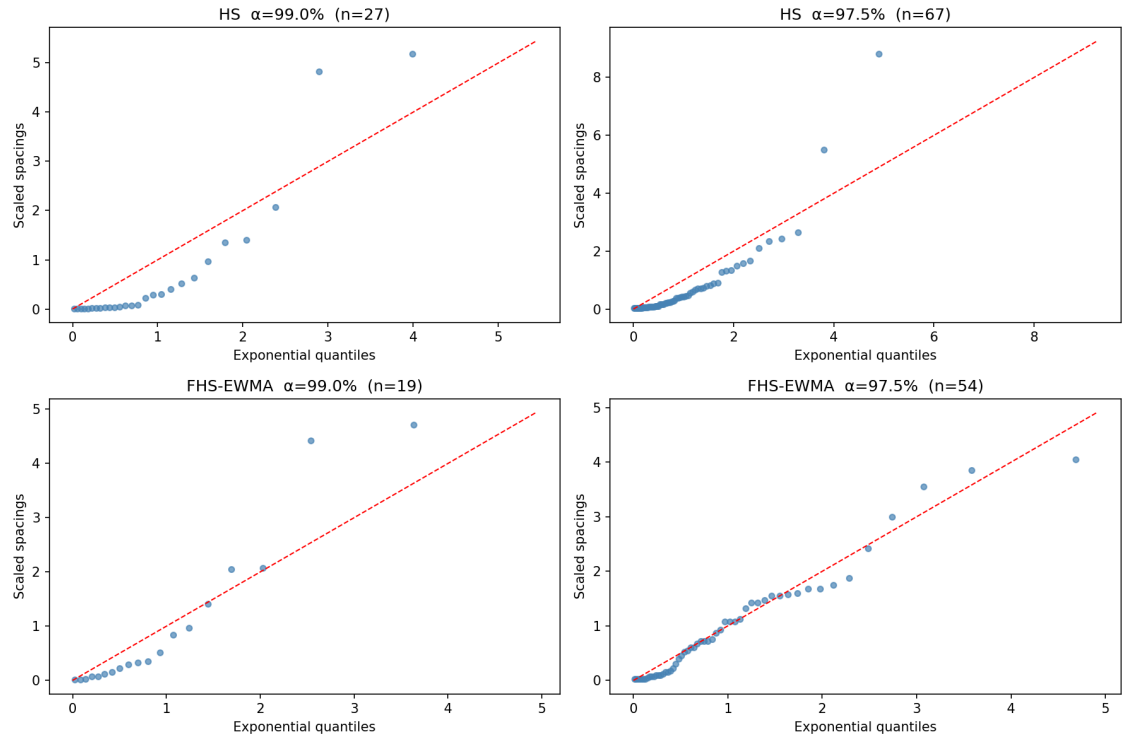


Figure 2.5: Q-Q plots of scaled violation spacings against standard exponential quantiles for HS (top row) and FHS-EWMA (bottom row) at $\alpha = 1\%$ (left) and $\alpha = 2.5\%$ (right).

Table 2.17: FHS yearly VaR violations at 1% and 2.5%.

Year	T	1%		2.5%	
		Expected	Actual	Expected	Actual
2017	6	0.06	0	0.15	0
2018	255	2.55	4	6.38	10
2019	255	2.55	3	6.38	8
2020	257	2.57	4	6.43	10
2021	258	2.58	1	6.45	7
2022	257	2.57	1	6.43	3
2023	255	2.55	0	6.38	3
2024	256	2.56	3	6.40	6
2025	255	2.55	3	6.38	7
2026	62	0.62	0	1.55	0
Total	2,120	21.20	19	52.93	54

Table 2.18: FHS-EWMA backtest: ES violation residual t -test.

α	n_{viol}	$\bar{K}$	t -stat	p -value	Pass
1%	19	0.006	0.117	0.908	Yes
2.5%	54	-0.026	-0.705	0.484	Yes

2.5.4 Model Comparison

Table 2.20 collects the elicibility scores across Models 3 and 4. A lower score indicates better VaR forecasts.

FHS-EWMA dominates HS on both the violation count and the elicibility score at both confidence levels. The lower score (783 vs 1014 at $\alpha = 1\%$) means the FHS VaR forecasts are on average closer to the realised quantile of the loss distribution. The improvement stems directly from conditioning on current volatility: during calm periods the VaR is trimmed, avoiding over-reservation, while during stress periods the forecast rises rapidly, reducing violations. HS, being unconditional, adjusts only as extreme events enter or exit the estimation window, making it structurally slower to respond.

2.6 Model 5: CCC-GARCH(1,1)

2.6.1 Method

The CCC GARCH(1,1) method allows the volatility $\sigma_{i,t}$ of each element i to vary according the following process

$$r_{i,t} = \mu + a_{i,t}, \quad a_{i,t} = \mu'_{i,t} + \sigma_{i,t} z_{i,t}, \quad z_t \sim \text{i.i.d. } N(0, 1)$$

and

$$\sigma_{i,t}^2 = \omega + \alpha a_{i,t-1}^2 + \beta \sigma_{i,t-1}^2,$$

where $r_{i,t}$ is the log-return on element i ($\omega > 0$, $\alpha \geq 0$, $\beta \geq 0$, and $\alpha + \beta < 1$ for covariance stationarity). We also note that we usually have $\mu'_{i,\cdot} = 0$ as there is weak predictability

Table 2.19: FHS-EWMA — average ES vs. realised shortfall on violation days (EUR).

Year	$\alpha = 1\%$			$\alpha = 2.5\%$		
	Violations	Avg ES	Avg Loss	Violations	Avg ES	Avg Loss
2017	0	—	—	0	—	—
2018	4	23,753	23,985	10	21,494	21,354
2019	3	27,403	23,373	8	19,229	17,710
2020	4	38,971	44,925	10	40,308	41,483
2021	1	28,397	28,142	7	23,537	19,886
2022	1	38,079	30,114	3	34,750	29,470
2023	0	—	—	3	25,168	20,477
2024	3	29,767	29,351	6	24,622	24,940
2025	3	40,641	44,206	7	27,509	30,256
2026	0	—	—	0	—	—
Total/Avg	19	32,148	32,878	54	26,975	26,307

Table 2.20: Elicitability VaR scores and backtest summary across models.

Model	Violations		Elic. score ($\times 10^3$)	
	$\alpha = 1\%$	$\alpha = 2.5\%$	$\alpha = 1\%$	$\alpha = 2.5\%$
HS	27	67	1 014	1 810
FHS-EWMA	19	54	783	1 538

in innovations. The constant conditional correlation CCC extension of GARCH assumes that the correlation $\rho_{i,j,t}$ between element i and j is constant over time, namely

$$\rho_{i,j,t} = \rho_{i,j}.$$

The method thus provides a time varying covariance matrix for the joint returns on the elements (we assume joint Gaussianity), where the change is fully driven by the volatilities through the GARCH(1,1) processes. We get the time-varying covariance matrix Σ_t through the following steps: **Step 1:** Estimate a univariate GARCH model for each element i :

$$\sigma_{i,t}^2 = \omega_i + \alpha_i \varepsilon_{i,t-1}^2 + \beta_i \sigma_{i,t-1}^2$$

Step 2: Construct the diagonal matrix of conditional standard deviations:

$$\Delta_t = \text{diag}(\sigma_{1,t}, \sigma_{2,t}, \dots, \sigma_{N,t})$$

Step 3: Standardize the demeaned return vector X_t :

$$Y_t = \Delta_t^{-1} X_t$$

Step 4: Estimate the constant correlation matrix:

$$P = \text{corr}(Y_t)$$

Step 5: Reconstruct the conditional covariance matrix:

$$\Sigma_t = \Delta_t P \Delta_t$$

. Through these 5 steps and according to the joint Gaussianity assumption, we can thus estimate the 1-day VaR and ES at level α and at time t for element i according to the equations

$$\text{VaR}_{\alpha,i,t} = -\mu_i + \sigma_{it}z_\alpha,$$

and

$$\text{ES}_{\alpha,i,t} = \mu + \sigma_{i,t} \frac{z_\alpha}{1 - \alpha}.$$

We also estimate the measures for the whole portfolio. Using the full sample to estimate the two risk measure, we can thus predict the 1-day VaR and ES by predicting the volatilities $\sigma_{i,t}$'s and the covariance matrix Σ_t through the CCC GARCH. We get the following results

Table 2.21: 1-day VaR and ES at 1% and 5% confidence levels (EUR).

Entity	VaR		ES	
	$\alpha = 1\%$	$\alpha = 5\%$	$\alpha = 1\%$	$\alpha = 5\%$
SHELL	2 888.28	2 032.88	3 377.05	2 620.80
ASML	8 804.49	6 177.02	10 440.21	8 117.30
JPM (EUR)	7 993.55	5 611.66	9 452.46	7 346.66
STOXX50	2 102.26	1 481.25	2 446.30	1 897.27
S&P 500 (EUR)	6 425.15	4 514.45	7 569.65	5 880.42
LOAN	942.87	669.43	1 059.88	818.14
Portfolio	20 582.67	14 425.79	24 522.72	19 073.83

2.6.2 Backtest

We backtest similarly to what was done for the previous methods and find the following results

Table 2.22: Yearly VaR and ES backtesting results

Year	Violations	Trading days	Expected	Avg Shortfall	Avg ES
2017	1	256	2.56	23 247.43	17 509.39
2018	7	255	2.55	23 432.16	21 738.08
2019	4	255	2.55	21 938.40	19 805.12
2020	10	257	2.57	39 274.45	35 670.41
2021	6	258	2.58	20 221.11	22 033.56
2022	6	257	2.57	26 257.45	29 967.83
2023	2	255	2.55	23 082.09	21 390.04
2024	6	256	2.56	24 940.08	22 544.00
2025	7	255	2.55	30 594.94	24 000.69
2026	0	62	0.62	0.00	24 541.36
Total	49	2 366	23.66	—	—

We now test for whether the model specification in this case (The multivariate Gaussian with CCC-GARCH volatility) is valid. For VaR testing, we perform the two sided test for $H_0 : I_c = \text{yearly number of violations} \sim \text{Bin}(T, 1 - \alpha_{\text{VaR}})$, where T is the number

of trading days in the specific calendar year and $\alpha_{\text{VaR}} = 99\%$. It is worth noting here that T is usually around 256, implying that the normal approximation for the binomial distribution is not valid as $T \times (1 - \alpha_{\text{VaR}}) = 256 \times (1\%) = 2.56 < 10$. We also test for the two sided null $H_0 : \text{ES} = \text{ES}_0$, to conclude with 95% confidence level whether the actual average shortfall observed is equal to the average yearly shortfall coming from the multivariate Gaussian model extended with CCC-GARCH(1,1). We implement a t-test by constructing a t-statistic using the mean violation residual. We get the following results.

Table 2.23: VaR and ES backtesting results

Year	VaR Violations	VaR CI	VaR Test	ES t-stat	ES CI	ES Test
2017	1	[0, 6]	Not rejected	-	[-12.7062, 12.7062]	Not rejected
2018	7	[0, 6]	Rejected	3.6242	[-2.3646, 2.3646]	Rejected
2019	4	[0, 6]	Not rejected	2.2088	[-2.7764, 2.7764]	Not rejected
2020	10	[0, 6]	Rejected	5.0209	[-2.2281, 2.2281]	Rejected
2021	6	[0, 6]	Not rejected	0.7951	[-2.4469, 2.4469]	Not rejected
2022	6	[0, 6]	Not rejected	-0.1002	[-2.4469, 2.4469]	Not rejected
2023	2	[0, 6]	Not rejected	-4.8345	[-4.3027, 4.3027]	Rejected
2024	6	[0, 6]	Not rejected	2.5589	[-2.4469, 2.4469]	Rejected
2025	7	[0, 6]	Rejected	2.9158	[-2.3646, 2.3646]	Rejected
2026	0	[0, 2]	Not rejected	-	[-2.3646, 2.3646]	Not rejected

We can observe that the VaR test is rejected in the years 2018, 2020 and 2025. The discrete nature of the binomial distribution makes it difficult to trust the results in years where $\hat{I}_c = 6$ as it lies in the right boundary of the confidence interval. However, the results of the expected shortfall testing are more reliable. The null is rejected in more years and is rejected in years where the VaR null is not. We can hypothesize that the expected shortfall null is rejected in years where volatility clustering patterns emerge. The following plot shows the volatility of the portfolio loss over the years.

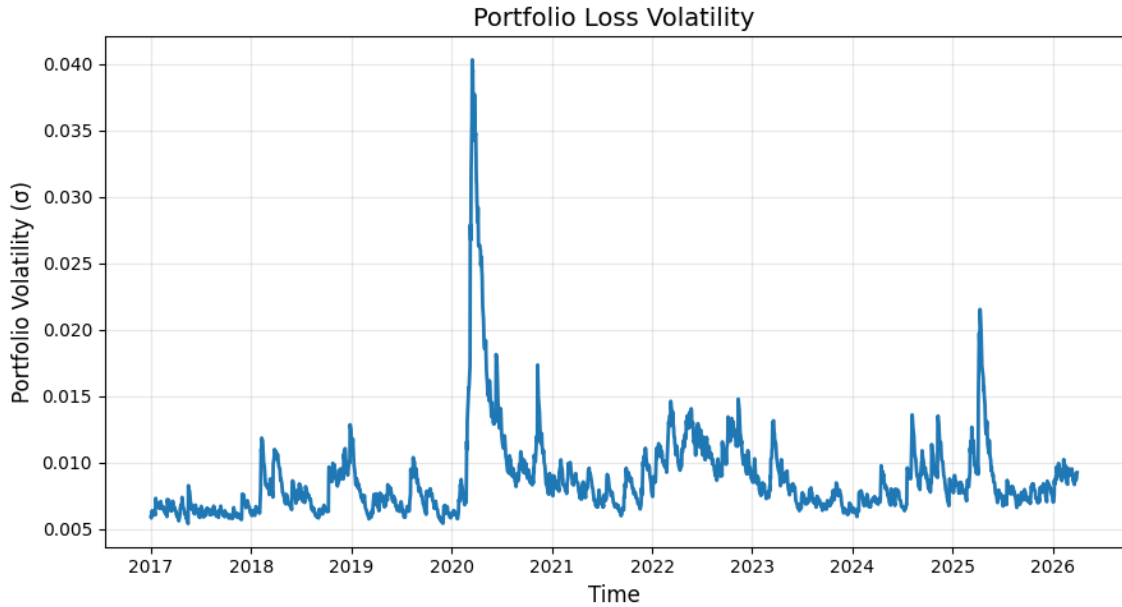


Figure 2.6: Portfolio loss volatility

Years such as 2018(start and end), 2020, 2024, and 2025 show volatility clusters. In 2022, however, there are patterns suggesting clustering, but the test was not rejected.

We can compare the results of table 2.22 with the results of the Gaussian model with constant volatility. The total number of violations is higher in the case of the CCC-GARCH extension ($n = 49$ vs $n = 46$). Furthermore, the CCC-GARCH extension captures the effect of volatility clustering, as violations are clustered and spread more around 2020 and 2022 compared to the constant volatility case. We can observe this through the following histogram plot.

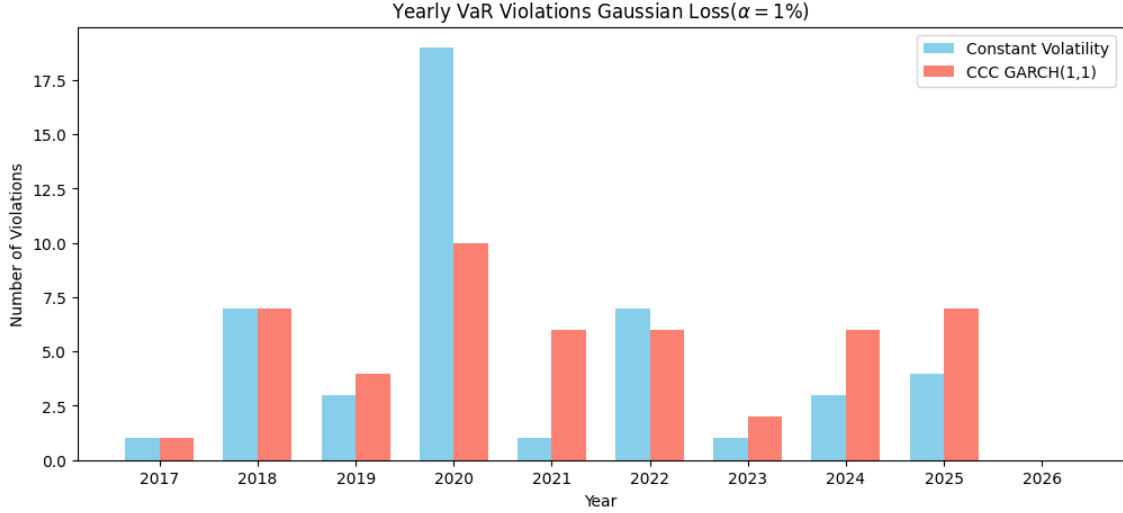


Figure 2.7: Number of violations under multivariate Gaussian model(σ vs σ_t)

We observe that when σ is constant, the model loads the number of violations on the most volatile year and unloads abruptly in the year after. This is different when volatilities follow a CCC-GARCH(1,1) process; Volatility clustering implies violation clustering in more volatile periods. This tells us that the portfolio risk from volatile periods is spread further in the future.

2.7 Task 6: Multi-day VaR

2.7.1 Method

The assignment requires 1-, 5-, and 10-day VaR to be constructed with the historical simulation method using *non-overlapping* returns at the corresponding horizon. Let $R_j^{(h)}$ denote the non-overlapping h -day portfolio simple return over block j , obtained by compounding daily simple returns:

$$R_j^{(h)} = \prod_{k=1}^h (1 + r_{j,k}) - 1,$$

where $h \in \{1, 5, 10\}$. The corresponding loss is

$$L_j^{(h)} = -V R_j^{(h)},$$

with $V = \text{EUR } 1,000,000$ the portfolio notional.

For each horizon, VaR and ES are then estimated from the empirical loss distribution using the same historical simulation estimators as in Model 3:

$$\widehat{\text{VaR}}_{\alpha}^{(h)} = L_{(K)}^{(h)}, \quad (2.7)$$

$$\widehat{\text{ES}}_{\alpha}^{(h)} = \frac{1}{n_h(1-\alpha)} \left[\sum_{k=K+1}^{n_h} L_{(k)}^{(h)} + (K - n_h\alpha)L_{(K)}^{(h)} \right], \quad (2.8)$$

where n_h is the number of non-overlapping observations at horizon h and $K = \lceil n_h\alpha \rceil$.

These direct multi-day estimates are compared with the square-root-of-time (SROT) approximation:

$$\widehat{\text{VaR}}_{\alpha, \text{SROT}}^{(h)} = \sqrt{h} \widehat{\text{VaR}}_{\alpha}^{(1)}.$$

This rule is exact only under restrictive assumptions such as iid returns and variance growing linearly with time. For a portfolio with compounding, fat tails and changing volatility, its accuracy is an empirical question.

2.7.2 Results

Table 2.24 reports the direct historical simulation estimates for the 1-, 5-, and 10-day horizons, together with the scaled 1-day VaR. The number of usable observations falls mechanically with the horizon because non-overlapping blocks are used: from 2 366 daily observations to 473 5-day blocks and 236 10-day blocks.

Table 2.24: Historical multi-day VaR and ES versus square-root-of-time scaling (EUR).

Horizon	α	n_h	HS VaR	HS ES	SROT VaR	Difference
1	99.0%	2 366	27 243	41 792	27 243	0
1	97.5%	2 366	19 724	30 394	19 724	0
5	99.0%	473	56 591	73 047	60 917	+4 325
5	97.5%	473	41 047	58 962	44 105	+3 058
10	99.0%	236	90 927	129 224	86 149	−4 777
10	97.5%	236	70 124	99 840	62 374	−7 750

At the 5-day horizon, the square-root-of-time rule slightly *overstates* VaR relative to direct historical simulation: by EUR 4 325 (7.6%) at the 99% level and EUR 3 058 (7.4%) at the 97.5% level. At the 10-day horizon, the picture reverses. The scaled VaR falls *below* the direct historical estimate at 99% by EUR 4 777 (−5.3%), and more materially at 97.5% by EUR 7 750 (−11.1%). Thus, the scaling rule is not uniformly biased; its error changes with the horizon and tail probability.

The ES figures increase more than proportionally with the horizon, especially at 10 days. At the 99% level, ES rises from EUR 41 792 at 1 day to EUR 73 047 at 5 days and EUR 129 224 at 10 days. This indicates that the shape of the multi-day loss distribution becomes increasingly heavy in the tail once compounding over longer horizons is taken into account.

Overall, the square-root-of-time rule provides a rough approximation for this portfolio over 5 days, but it is clearly less reliable over 10 days. This is not surprising: the rule ignores compounding, fat tails and volatility clustering, all of which are present in the data and already visible in the backtests of the 1-day models. For risk management purposes,

the direct historical multi-day estimate is therefore preferable whenever sufficient data are available.

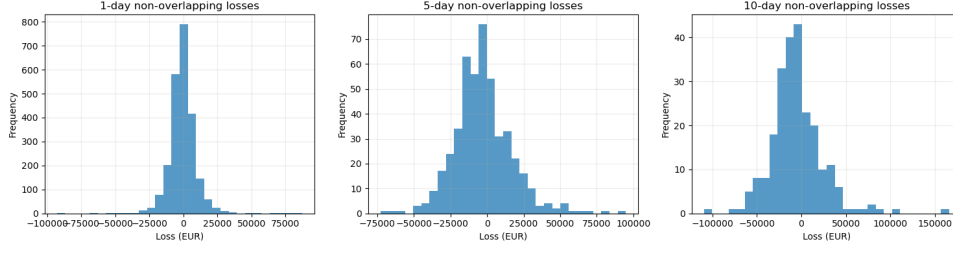


Figure 2.8: Historical simulation multi-day VaR compared with square-root-of-time scaling.

2.8 Task 7: Stress Testing

2.8.1 Method

The assignment asks for explicit single-factor stress scenarios applied to the *current* portfolio. We therefore shock the portfolio at the end of the sample (March 2026), one factor at a time, and revalue the current euro notionals under each scenario. The base 1-day historical VaR at that date is EUR 27 243 and the corresponding ES is EUR 41 792.

The following stress scenarios are considered:

- Equity and index prices: $\pm 20\%$ and $\pm 40\%$
- FX (major currency): $\pm 10\%$ in EUR/USD
- Interest rate: ± 2 and ± 3 percentage point shifts in EURIBOR

Commodity shocks are not implemented because the portfolio has no direct commodity exposure, so their effect is zero by construction.

For equities and indices, the shocked notional is obtained by multiplying the current euro notional by $(1 + s)$ where s is the scenario shock. For the EUR/USD stress, only the USD-origin positions (JPM and S&P 500) are affected. Because EUR/USD is quoted as USD per EUR, the euro value of USD assets scales as

$$\text{EUR value multiplier} = \frac{1}{1 + s}.$$

Thus a EUR/USD increase reduces the euro value of USD assets, while a decrease raises it.

For the loan, the immediate mark-to-market effect of an interest-rate shock Δy is approximated using modified duration:

$$\frac{\Delta P}{P} \approx -D^* \Delta y,$$

with $D^* = 9.135$ from the portfolio construction section.

After the instantaneous revaluation, a stressed 1-day historical VaR and ES are computed by applying the historical daily return scenarios to the stressed euro notionals. This isolates how a change in today's portfolio composition affects the next-day empirical loss distribution.

2.8.2 Results

Table 2.25 reports the immediate mark-to-market effect and the stressed 1-day historical VaR and ES for the main scenarios.

Table 2.25: Stress test results: immediate P&L and stressed 1-day HS VaR/ES (EUR).

Factor	Shock	Instant P&L	Stressed VaR	Stressed ES	Δ VaR
ASML	−40%	−64 000	24 423	38 475	−2 820
ASML	−20%	−32 000	25 378	40 044	−1 865
ASML	+20%	32 000	28 737	43 614	1 495
ASML	+40%	64 000	30 020	45 446	2 777
Shell	−40%	−32 000	26 625	39 974	−618
Shell	−20%	−16 000	26 971	40 872	−272
Shell	+20%	16 000	27 515	42 737	272
Shell	+40%	32 000	28 075	43 690	832
JPM (EUR)	−40%	−96 000	23 857	36 191	−3 386
JPM (EUR)	−20%	−48 000	25 834	38 987	−1 409
JPM (EUR)	+20%	48 000	28 693	44 643	1 450
JPM (EUR)	+40%	96 000	30 117	47 616	2 874
EURO STOXX 50	−40%	−32 000	26 119	40 421	−1 123
EURO STOXX 50	−20%	−16 000	26 835	41 100	−408
EURO STOXX 50	+20%	16 000	27 804	42 489	562
EURO STOXX 50	+40%	32 000	28 366	43 186	1 123
S&P 500 (EUR)	−40%	−96 000	24 794	37 649	−2 448
S&P 500 (EUR)	−20%	−48 000	25 829	39 701	−1 414
S&P 500 (EUR)	+20%	48 000	28 713	43 884	1 470
S&P 500 (EUR)	+40%	96 000	29 831	45 998	2 589
EUR/USD	−10%	53 333	28 837	44 530	1 595
EUR/USD	+10%	−43 636	25 960	39 567	−1 283
EURIBOR	−3pp	54 811	27 193	41 798	−49
EURIBOR	−2pp	36 540	27 178	41 795	−65
EURIBOR	+2pp	−36 540	27 308	41 788	65
EURIBOR	+3pp	−54 811	27 340	41 787	98

Several patterns stand out. First, the largest immediate mark-to-market losses come from −40% shocks to the largest risky positions, namely JPM and the S&P 500, each reducing portfolio value by EUR 96 000. ASML follows with a EUR 64 000 loss under a −40% scenario. This simply reflects the portfolio weights: the larger the position, the larger the instantaneous impact.

Second, the effect on *VaR* is not the same as the immediate P&L effect. A negative shock lowers the future euro notional invested in that asset and therefore often *reduces* next-day VaR. For example, a −40% shock to JPM reduces stressed VaR to EUR 23 857, i.e. EUR 3 386 below the base VaR. Conversely, a +40% shock to the same position raises stressed VaR to EUR 30 117, the highest value across all scenarios and EUR 2 874 above the base level. The same mechanism appears for ASML and the S&P 500.

Third, FX shocks affect only the USD-linked positions. A 10% fall in EUR/USD increases the euro value of these positions and raises VaR to EUR 28 837, whereas a 10% rise in

EUR/USD lowers their euro value and reduces VaR to EUR 25 960. This is exactly what we would expect from the EUR translation mechanics.

Finally, the interest-rate stresses have only a modest effect on VaR and ES. Even a ± 3 percentage point EURIBOR shock moves VaR by less than EUR 100. The reason is that the loan represents only 20% of the portfolio, and the stress is applied to a single fixed-rate instrument rather than to the whole risky sleeve. The immediate P&L effect of the duration shock is visible, but its contribution to next-day portfolio VaR remains limited relative to the equity exposures.

Overall, the findings match prior expectations. Equity shocks dominate both the instantaneous mark-to-market effect and the change in stressed VaR. Currency shocks matter because two of the major risky exposures originate in USD. Interest-rate shocks are economically meaningful for the loan itself, but they do not dominate total portfolio risk because the fixed-income sleeve is smaller than the combined equity sleeve.

2.8.3 Worst-case Scenarios

The most severe immediate losses occur for the -40% shocks to JPM and the S&P 500 (both EUR $-96\,000$), followed by the -40% ASML shock (EUR $-64\,000$) and the $+3$ percentage point rate shock on the loan (EUR $-54\,811$). By contrast, the highest stressed VaR occurs under *positive* shocks to the large equity positions: JPM $+40\%$ raises VaR to EUR 30 117, ASML $+40\%$ to EUR 30 020, and the S&P 500 $+40\%$ to EUR 29 831. This distinction is important from a risk-management perspective: the scenario that hurts the portfolio *today* is not necessarily the one that maximises the portfolio's *future* one-day VaR.

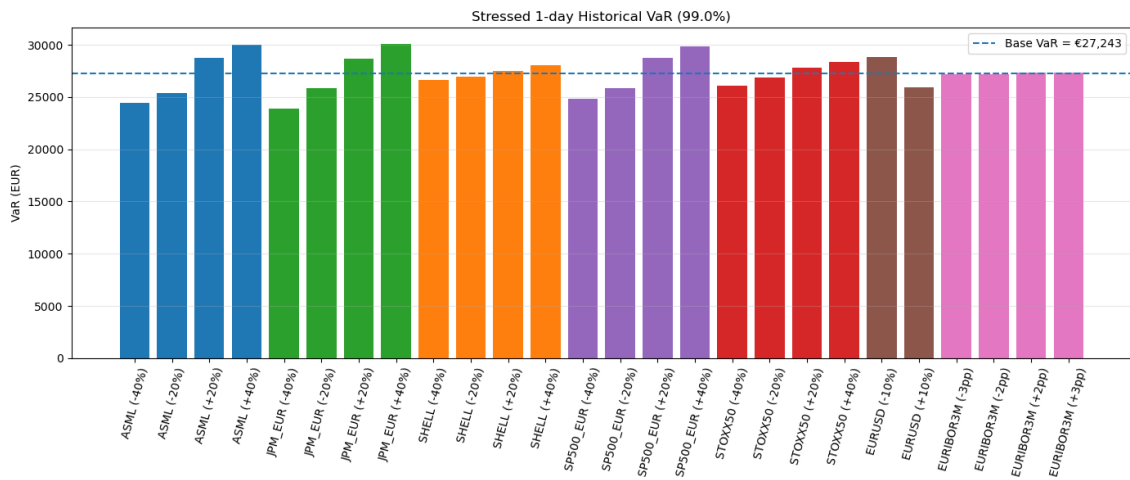


Figure 2.9: Stressed 1-day historical VaR across single-factor shock scenarios. The dashed line denotes the base VaR of EUR 27 243.

3 CONCLUSIONS AND RECOMMENDATIONS

3.1 Model Performance

The method based on the normal distribution yields the weakest performance. It underestimates tail risk and the volatility clustering present in certain years, leading to a high number of violations. The Student-t model performs better on this front, due to the heavier tails reducing the gap in VaR violations and ES underestimation. The performance works well in normal years, but suffers similar to the normal model under high stress years.

The CCC-GARCH model introduces time-varying volatility but produces the most violations overall. It is, however, more consistent by spreading out violations evenly throughout years.

The historical simulation method is an improvement over the normal and student-t models, but has the same weaknesses – it does not react to changing volatility with enough speed so clustering persists.

The FHS-EWMA/ Filtered Historical Simulation is the best performing model. It has a fast reacting volatility filter which causes VaR to spike in stressful periods. Violations are very close to expected amounts.

3.2 Recommendations

The FHS-EWMA model should be used as the primary model for risk measurement. A more advanced GARCH model like the DCC-GARCH should also be looked into.

The square root of time rule - SROT rule - is unreliable beyond 5 days and is not preferred. Combined stress tests would be useful, as current scenarios only look at singular shocks.

The current portfolio was optimised over a pre-sample period, and the weights maintained. The estimated covariance matrices would decay in quality as time passes, and hence the portfolio can also be re-optimised periodically to decrease risk. There were also no hedge assets involved, in practise FX risk and interest rate risk can be hedged using derivative instruments like FX and IR swaps.

APPENDIX



Figure 3.1: Prices of Assets

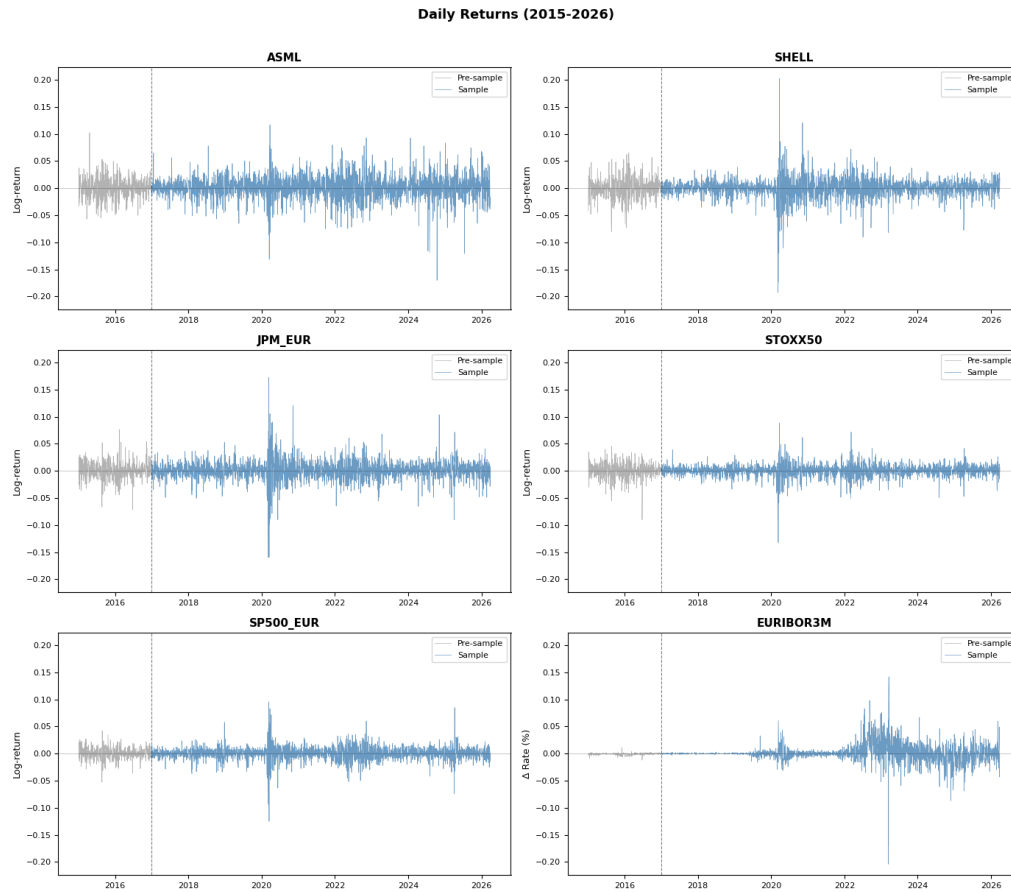


Figure 3.2: Log Returns of Assets

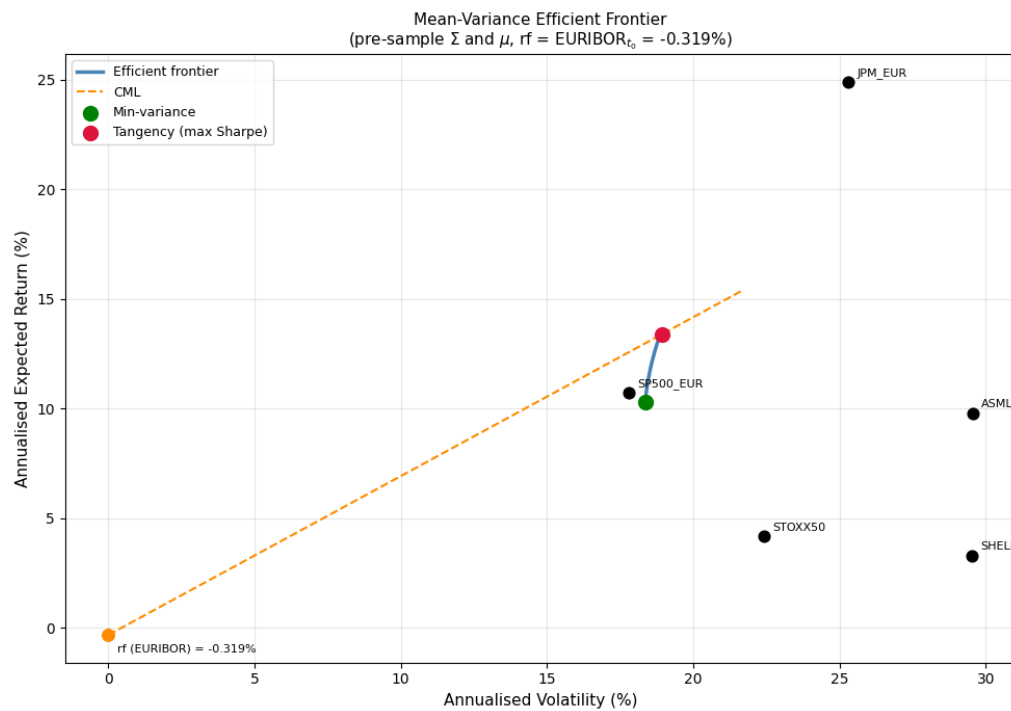


Figure 3.3: Efficient Frontier